Free Radical Research

ISSN: 1071-5762 (Print) 1029-2470 (Online) Journal homepage: http://www.tandfonline.com/loi/ifra20

Ergothioneine and melatonin attenuate oxidative stress and protect against learning and memory deficits in C57BL/6J mice treated with D-galactose

T.-Y. Song, H.-C. Lin, C.-L. Chen, J.-H. Wu, J.-W. Liao & M.-L. Hu

To cite this article: T.-Y. Song, H.-C. Lin, C.-L. Chen, J.-H. Wu, J.-W. Liao & M.-L. Hu (2014) Ergothioneine and melatonin attenuate oxidative stress and protect against learning and memory deficits in C57BL/6J mice treated with D-galactose, Free Radical Research, 48:9, 1049-1060, DOI: 10.3109/10715762.2014.920954

To link to this article: http://dx.doi.org/10.3109/10715762.2014.920954

Accepted author version posted online: 06 May 2014.
Published online: 21 Jul 2014.

Submit your article to this journal

Article views: 169

View related articles

View Crossmark data

Citing articles: 12 View citing articles

Full Terms & Conditions of access and use can be found at
http://www.tandfonline.com/action/journalInformation?journalCode=ifra20

Download by: [University of Sussex Library] **Date:** 24 September 2017, At: 10:05

Free Radical Research, September 2014; 48(9): 1049–1060
© 2014 Informa UK, Ltd.
ISSN 1071-5762 print/ISSN 1029-2470 online
DOI: 10.3109/10715762.2014.920954

informa
healthcare

ORIGINAL ARTICLE

Ergothioneine and melatonin attenuate oxidative stress and protect against learning and memory deficits in C57BL/6J mice treated with D-galactose

T.-Y. Song[1], H.-C. Lin[1], C.-L. Chen[1], J.-H. Wu[2], J.-W. Liao[2] & M.-L. Hu[3]

[1]*Department of Health Food, Chung Chou University of Science and Technology, Changhua, Taiwan,* [2]*Graduate Institute of Veterinary Pathobiology, National Chung Hsing University, Taichung, Taiwan, and* [3]*Department of Food Science and Biotechnology, National Chung Hsing University, Taichung, Taiwan*

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Abstract

Male C57BL/6J mice treated with D-galactose (DG) were used to examine the effects of ergothioneine (EGT), melatonin (MEL), or their combination (EGT+MEL) on learning and memory abilities. The mice were divided into five groups and injected subcutaneously with DG (0.3 mL of 1% DG/mouse) except for group 1 (normal controls). Group 3 was orally supplemented with EGT [0.5 mg/kg body weight (bw)], group 4 with MEL (10 mg/kg bw, p.o.), and group 5 with EGT+MEL. EGT and MEL were provided daily for 88 days, while DG was provided between days 7 to 56. Active avoidance task and Morris water-maze task were used to evaluate learning and memory abilities. DG treatment markedly increased escape latency and decreased the number of avoidance in the active avoidance test, whereas EGT and MEL alone significantly improved the performance. DG also impaired the learning and memory abilities in the water-maze task, and EGT and MEL alone also significantly improved the performance. EGT+MEL produced the strongest effects in both tasks. EGT and MEL alone markedly decreased β-amyloid protein accumulation in the hippocampus and significantly inhibited lipid peroxidation and maintained glutathione/glutathione disulfide ratio and superoxide dismutase activity in brain tissues of DG-treated mice. MEL alone completely prevented the rise in brain acetylcholine esterase activity induced by DG, whereas EGT and EGT+MEL were only partially effective. Overall, EGT, MEL, and, in particular, the combination of EGT and MEL effectively protect against learning and memory deficits in C57BL/6J mice treated with DG, possibly through attenuation of oxidative damage.

Keywords: ergothioneine, melatonin, D-galactose, Alzheimer's disease, learning and memory

Abbreviations: AChE, acetylcholinesterase; AD, Alzheimer's disease; Aβ, β-amyloid protein; DG, D-galactose; EGT, ergothioneine; MEL, melatonin; OCTN, organic cation transporter; TBARS, thiobarbituric acid-reactive substances

Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disease of the brain that is characterized by deterioration of memory and cognitive function leading to dementia, unusual behavior, personality changes, and ultimately death [1]. Deposition of β-amyloid protein (Aβ) in the brain is known to increase oxidative stress leading to neurodegeneration and disrupted cognitive function [2,3]. Oxidative stress generated from Aβ deposition in the brains of AD patients has been implicated as an important factor implicated in AD [4–6]. Indeed, oxidative stress and Aβ production are proportionally linked to each other because the former can induce the latter and vice versa both *in vivo* and *in vitro* [7].

Ergothioneine (2-mercaptohistidine trimethylbetaine; EGT), which is formed in some bacteria and fungi but not in animals [8], is known to be present in mammal brain *in vivo* at the level of 0.2–1.0 mg per 100 g tissue [9]. In humans, EGT is only absorbed through consumption of plant diet, primarily by consumption of edible mushrooms. *In vitro* studies have shown that EGT possesses antioxidant, anti-radiation, and inflammatory activities [10,11], and much evidence suggests that EGT may function as a physiological antioxidant [See ref. 11 for a review]. EGT also protects neurons from cytotoxicity induced by a variety of neurotoxins, including N-methyl-D-aspartate, β-amyloid, and cisplatin [12–14]. In addition, EGT confers cellular homeostasis in neuronal cells challenged with the pro-oxidant mixture of *N*-acetylcysteine and hydrogen peroxide [15]. However, in spite of all the evidence about their beneficial effects, little is known whether EGT is able to improve parameters related to oxidative stress that are altered by aging.

Melatonin (N-acetyl-5-methoxytryptamine; MEL) is the main hormone of the pineal gland that has been described to be a potent antioxidant [16,17]. Studies have shown that MEL levels are lower in AD patients compared with those in age-matched control subjects [18,19]. MEL has a multitude of other functions that could support most of the AD hypotheses such as Aβ accumulation, cholinergic, neuroinflammation, calcium, insulin resistance, and

Correspondence: Miao-Lin Hu, Ph.D, Department of Food Science and Biotechnology, National Chung Hsing University, 250 Kuo Kuang Road, Taichung, 402 Taiwan. Tel: + 886-4-2281-2363. Fax: + 886-4-2281-2363. E-mail: mlhuhu@nchu.edu.tw.

(Received date: 15 January 2014; Accepted date: 30 April 2014; Published online: 2 June 2014)

peroxidation of brain lipids hypothesis [20]. In this context, we recently showed that MEL ameliorates the loss of learning and memory in Aβ-treated mice, possibly through its antioxidant activity [21].

D-Galactose (DG) is normally metabolized by galactose mutarotase, galactokinase, galactose-1-phosphate uridyltransferase, and UDP-galactose epimerase in the human body [22]. Overdose of DG beyond the capacity of those two enzymes allows aldose reductase to catalyze accumulated DG into galactitol, which cannot be metabolized but will accumulate in the cell, leading to osmotic stress and generation of ROS [23]. Chronic systemic exposure of animals to DG leads to the acceleration of senescence, which is very similar to the natural aging in animals, such as a shortened lifespan [24], cognitive deficits [25], neurologic impairment associated with the increases of free radicals and the decreases of antioxidant enzyme activity [24,25] and immune responses [26]. Therefore, by injecting DG into rats or mice for over a month has been widely used in recent years as an aging model [27–31]. Using DG-induced Aβ formation model in C57BL/6J mice, we have previously demonstrated that oral supplementation with isoflavone attenuates oxidative stress and improves parameters related to aging and AD in C57BL/6J mice treated with DG [30]. However, it is unclear whether DG may induce learning and memory deficits in mice.

Many attempts have been made to improve cognitive deficits by increasing brain cholinergic activity with acetylcholinesterase (AChE) inhibitors such as donepezil. However, the resulting side effects such as pain, nausea, and vomiting associated with these agents have limited their use [32,33]. Therefore, it is worthwhile to explore the utility of phytochemicals or the combination of a phytochemical with a drug for the treatment of various cognitive disorders. In the present study, we investigated the neuroprotective effects of EGT, MEL, and their combination using the DG-induced learning and memory deficits and brain oxidative damage in C57BL/6 mice.

Materials and methods

Chemicals

Chemicals including EGT, D-galactose, n-butanol, thiobarbituric acid, 1,1,3,3-tetraethoxypropane, and all other reagents were purchased from Sigma Chemical Company (St. Louis, MO, USA). The β-amyloid, 17-24 (4G8) mouse monoclonal antibody was provided by Covance (SIG-39220), and OCTN1 Antibody (C-13) goat polyclonal antibody was provided by Santa Cruz Biotechnology, Inc. (sc-19819).

Animal experimental design

C57BL/6 mice (27 weeks old) weighing about 30 ± 2 g were used for the experiments. The mice were maintained on a 12:12-hour light: dark cycle and provided with food (LabDiet 5001 Rodent Diet; PMI Nutrition International) and water *ad libitum* throughout the study (a total of 88 days). EGT and MEL were dissolved in normal saline and supplemented orally (p.o.) at 0.5 mg and 10 mg/kg bw, respectively, starting from day 1 till day 88. DG was administered subcutaneously (s.c., 0.3 mL of 1% solution dissolved in normal saline. i.e., 120 mg DG/mouse) [18] daily between days 7 and 56. The mice were divided into five groups (9 mice per group) as follows: group 1: control; group 2: DG + saline (p.o., 10 μL/kg bw to replace EGT or MEL); group 3: DG+EGT; group 4: DG+MEL; group 5: DG+EGT+MEL. The doses of EGT and MEL were the same as those used in our previous studies [21]. The EGT dose (0.5 mg/kg/day) is equivalent to a human consumption level of 2.5 mg/day, which can easily be achieved by consumption of EGT-rich foodstuff, such as mushrooms (an intake of approximately 4 g dry mushroom per day would achieve such an intake) [21]. The experimental scheme is shown in Figure 1. All experimental procedures involving animals were conducted in accordance with National Institutes of Health (NIH) guidelines.

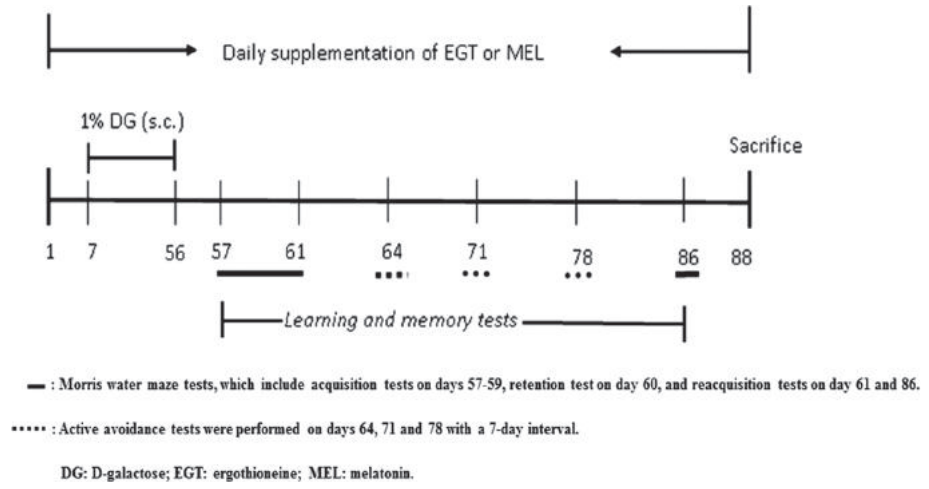

Figure 1. Experimental scheme for the treatment of D-galactose (DG), ergothioneine (EGT), melatonin (MEL), and EGT+MEL as well as the active avoidance tests and water maze tests.

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

This experiment was approved by the Institutional Animal Care and Use Committee (IACUC, no. 016) of the Chung Chou University of Technology.

Active avoidance test

The active avoidance test in a shuttle box was used to evaluate the memory and learning ability using a Gemini Avoidance System (San Diego Instruments, San Diego, CA, USA), as described previously [34] with minor modification. The test was measured as step-through latency and number of successful avoidance, and the shorter latency and higher avoidance number the mouse escapes, the better learning and memory the mouse has. Each mouse was given five trials daily with an inter-trial interval of 15 s. On the first trial, mice were placed individually into the large compartment of the apparatus with the door opened and accommodated therein for about 10 s. A light bulb (60 W) was switched on randomly in the opposite compartment of mice was located. The light stimulus which preceded the onset by 5 s of an electric shock (0.3 mA for 5 s) applied to the floor grid. The light stimulus is unrelated to the electric shock, but if the light is paired multiple times with the electric shock, the light would eventually trigger the response, that is, to move the mice to the light room. If the mice ran into the other (light) compartment within 5 s after the onset of the light stimulus, the microprocessor recorder unit of the shuttle box would record an avoidance number.

The step-through latency was the time required for a mouse to move to the other compartment after the onset of the light stimulus, and a 'no response' was recorded, if a mouse had no response for a maximum escape latency of 15 s after the onset of the light stimulus. The step-through latency and number of successful avoidance were measured on days 64, 71, and 78 (with a 7-day interval) as formal records [21]. The recorder unit of automated shuttle box continuously recorded this parameter during all experimental periods.

Water-maze task

The Morris water-maze task, conducted according to the method described previously [34] with some modification, was used to evaluate the spatial memory and learning abilities that are referred to as the acquisition test, retention test, and reacquisition test, as described separately below. A stainless steel circular tank (diameter 160 cm, height 60 cm) was used as the apparatus of the water maze in which a movable escape platform (diameter 9 cm, height 26 cm) was hidden inside the tank. The tank was filled to a height of 28 cm with water of approximately 23 ± 2°C that was clouded with powdered milk. The circular tank was divided into four quadrants (R1, R2, R3, and R4), and a position with equal distance from center and edge in the middle of each quadrant was marked for the location of platform. The water tank was located in a test room with many cues external to the maze. The room had adjustable indirect light, and a camera was set on the ceiling above the center of water tank, and the record program was the SINGA Real-Time Trace Mouse Ver. 1.17 System (Taipei, Taiwan). The position of the cues remained unchanged throughout the water maze task.

Water-maze acquisition test

Acquisition refers to the first stage of learning, when a response is established. In this test (conducted four times a day on days 57 to 59), a transparent platform was hidden in a constant position of tank (R2 quadrant) throughout the test period, and its top was equal to the water surface in R2 quadrant of the maze. For each trial, the mouse was put into the pool in one of the four quadrants, with the sequence of the positions being selected randomly. In each trial, the latency to escape on to the hidden platform was recorded. If the mouse found the platform, it was allowed to remain there for 30 s and was then returned to its home cage. If the mouse was unable to find the platform within 120 s, the training was terminated, and a maximum score of 120 s was assigned. Training was conducted for 3 consecutive days, four times a day, from days 57 to 59 (a total of 12 trials). By means of the 12 trials of acquisition tests, mice gradually recognized the location of escape platform to display the reduction of escape latency.

Water-maze retention test

Retention test is to determine whether the mice have obtained the memory after the acquisition test. This test (conducted on day 60) is different from the acquisition test in that the hidden transparent platform was removed from the pool. Each mouse was put into the pool from the R4 quadrant and allowed to swim for 60 s in the water maze. The test was carried out by recording the swimming route for each mouse in the pool and calculating the swimming distance and time to cross the R2 quadrant where the hidden platform was located in the acquisition test but was removed in this test.

Water-maze reacquisition test

A reacquisition test (conducted on days 61 and 86) is a learning ability test that determines whether the mice have established a new strategy to find a hidden transparent platform which is submerged into the water (0.4 cm) in a different quadrant (i.e., at R3) from that (at R2) conducted in the acquisition test. The mouse was first put into the pool from the R1 quadrant and trained three times to find the hidden platform in the new quadrant (R3) within 120 s. The formal test (working memory test) was performed 4 h after training by placing the mouse in the R2, instead of R1, quadrant of the maze. The latency to escape on to the hidden platform (at R3 quadrant) was recorded, and the training was terminated and a maximum score of 120 s was assigned.

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Preparation of brain tissue homogenates

After completion of treatment with EGT (on day 88), the mice were killed by decapitation under CO_2 anesthesia. All following procedures were carried out at 0–4°C. Whole brains, except cerebellum, were homogenized using an Ultra Turax homogenizer in 1:10 (w/v) buffer (5.0 mM Tris base, 150 mM NaCl, and 20 mM EDTA, pH 7.5). The homogenates were sonicated for 30 s in a disruptor (Bronson Sonic, NY, USA) and centrifuged at 16,000 × g for 10 min. The supernatant was kept at −70°C until use.

Determination of GSH/GSSG ratio in brain tissues

GSH/GSSG ratios in brain tissue homogenates were assayed according to the HPLC method of Schofield and Chen [35] with some modification. In brief, after adding 100 μL of 10% perchloric acid to precipitate the protein of brain homogenates (1.0 mL), the supernatants were treated with iodoacetic acid in the dark at 40°C for 1 h. A volume (0.2 mL) of 3% (v/v) 2,4-dinitrofluorobenzene was added to the reaction mixture and allowed to react at 40°C for 4 h in the dark. After centrifugation (3000 g for 15 min), a portion (25 μL) of the supernatant was injected onto HPLC column. The HPLC system consisted of a Hitachi l-6200 pump, a Hitachi D-2500 chromatointegrator, a Hitachi L-4200 UV-Vis detector, and a 3-aminopropyl column (200 × 4 mm, 5 μm). The mobile phase consisted of (A) water/methanol (1:4, v/v) and (B) 2 M sodium acetate, pH 4.6/methanol (36:64, v/v). The 3-aminopropyl column was eluted by a 30-min gradient from 75% (A) to 5% and then isocratically maintained for 15 min at a flow rate of 1.2 mL/min and UV detection at 365 nm.

Determination of thiobarbituric acid-reactive substances in brain tissues

Lipid peroxidation in brain tissue homogenates was determined using the thiobarbituric acid (TBA) assay [36]. Butylated hydroxytoluene (10 μL, 50 mM) was added to the tissue homogenate (1.0 mL) to terminate the peroxidation reaction and then mixed with 1 mL of 7.5% (w/v) cold trichloroacetic acid (TCA) to precipitate proteins. The supernatant was allowed to react with 1 mL of 0.8% (w/v) TBA in a boiling water bath for 45 min. After cooling, levels of TBARS were determined spectrofluorimetrically (at 515-nm excitation and 555-nm emission) using 1,1,3,3-tetraethoxypropane as the standard and were expressed as nmol per milligram of protein.

Determination of AChE activity and EGT contents in brain tissues

AChE activity in brain tissues was measured by the spectrophotometric method as developed by Ellman et al. [37]. For determination of EGT contents, we used the method outlined in detail by Dubost et al. [38] for measuring EGT in mushrooms. In short, analysis was carried out using an HPLC with separation carried out on two Econosphere C18 columns (Thermo scientific Associates, IL, USA) with each column being 250 mm × 4.6 mm, 5 μm particle size connected in tandem. The isocratic mobile phase was 50 mM sodium phosphate in water with 3% acetonitrile and 0.1% triethylamine adjusted to a pH of 7.3 with a flow rate of 1 ml per minute. The injection volume was 20 μl, with the column temperature being ambient. EGT was quantified by monitoring absorbance at 254 nm using the authentic standard. Data were expressed as mmol of EGT per milligram of protein (mmol/mg protein). Protein concentrations were determined using a standard commercial kit (Bio-Rad Laboratories Ltd., USA) with serum bovine albumin as standard.

Immunohistochemical staining of Aβ and OCTN1 in hippocampus

The brain was fixed in 10% formalin at pH 7.4. Brains tissues were blocked, and serial 2.5-μm-thick frozen sections cut on a sledge microtome were collected sequentially and without interruption into wells. The tissue was dried at 60°C for 1 h, then deparaffinized and hydrated, following a dip of xylene, gradient concentration of ethanol, and PBS. These sections were treated for 30 min in 3% hydrogen peroxide in absolute methanol, followed by several washes with cold PBS. After blocking in 10% normal goat serum in PBS, sections from regions containing hippocampus were processed for Aβ and OCTN1 immunohistochemical staining by placing them in primary monoclonal Aβ1-40 or OCTN1 antibody (1:100) diluted in antibody diluents at 4°C overnight. Labeling was detected by using the horseradish peroxidase (HRP) and peroxide/DAB immunohistochemistry detection kit (Dako North America, Inc., Carpinteria, CA, USA). Counterstaining of the section was then carried out with hematoxylin for 3 min and stopped with a PBS wash. Sections were then mounted, air dried, and coverslipped. The Aβ1-40 and OCTN1 accumulation in hippocampus was monitored by intermittent microscope examination and digital image system (Image Pro Plus 6.2) provided by Total Smart Technology Co., Ltd. (Taipei, Taiwan). The percentage of Aβ accumulation was expressed as:

$$\frac{\text{(total A}\beta\text{ or OCTN1 plaque area in cover slipped)}}{\text{(total cortex area in cover slipped)}} \times 100\%.$$

Statistical analysis

All statistical analyses were performed using SPSS for Windows, version 17 (SPSS, Inc., Chicago, IL, USA). Data are expressed as means ± SD and analyzed using one-way ANOVA followed by Duncan's multiple range test. $P < 0.05$ is considered statistically significant.

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Results

Body weight changes

Body weights of mice were measured once a week at weeks 1, 4, 8, 12 during the 88-day study period. DG treatment alone tended to decrease the body weight of mice at weeks 8 and 12, as compared to the control mice ($P > 0.05$). Supplementation with EGT (0.5 mg/kg bw), MEL (10 mg/kg bw) or EGT+MEL appeared to prevent the body weight loss induced by DG treatment. However, none of these changes were statistically significant (data not shown).

Active avoidance test

Active avoidance test was performed on days 64, 71, and 78. We found no significant differences in escape latency between each treatment group on day 64 (Figure 2A). However, the escape latency of each treatment group was decreased with the increase in the number of training. The results obtained on days 71 and 78 show that DG alone significantly increased the step-through latency, as compared to the control group. Supplementation with MEL alone significantly decreased the step-through latency both on days 71 and 78, whereas supplementation with EGT alone or EGT+MEL only significantly decreased the step-through latency on day 78, as compared to DG treatment alone. No significant differences existed in the step-through latency among the three supplemental groups of mice on day 78, as the step-through latencies of these supplemental groups had reached the level of the normal control group.

The number of avoidance in mice treated with DG alone was significantly lower than that of control group ($P < 0.05$) on day 78, but not on days 64 and 71 (Figure 2B). Neither EGT nor MEL treatment alone significantly increased the number of avoidance on day 78, whereas EGT+MEL significantly increased the number of avoidance, as compared to DG treatment alone, to an extent that was not significantly different from the number of avoidance in the normal control group ($P < 0.05$) (Figure 2B).

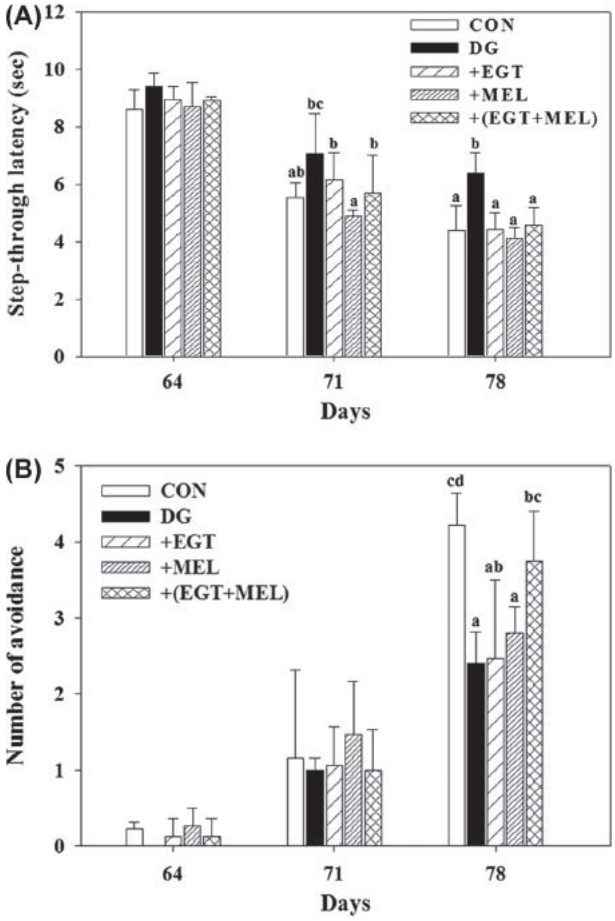

Figure 2. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on step-through latency (A) and number of successful avoidance (B) in the active avoidance tests on days 64, 71, and 78 in mice treated s.c. with D-galactose (DG, 1%). The DG-treated mice were supplemented p.o. with EGT (0.5 mg/kg/d), MEL (10 mg/kg/d), or EGT+MEL. Values not sharing an alphabetic letter are significantly different ($P < 0.05$).

Water-maze test

Water-maze acquisition test. Learning-related behaviors in a water maze were performed immediately following the termination of DG treatment (days 7–56) with the acquisition tests carried on days 57 to 59 (a total of 12 tests). As shown in Figure 3A, the escape latency (i.e., time spent to find a hidden platform at R2 quadrant) was gradually decreased with the increasing number of test in each treatment group (even in DG-treated group). Even with this improvement, the differences in escape latencies between the DG treatment group and the other groups became pronounced after the fourth test (four tests a day). In the last four tests (conducted on day 59), EGT or MEL treatment tended to decrease the escape latency, but only the effect of EGT+MEL was significantly different from that of DG treatment alone ($P < 0.05$). Figure 3B shows that mice treated with DG alone searched the target platform in a directionless manner and swam around the entire pool. In contrast, mice supplemented with MEL, EGT, or EGT+MEL decreased the escape latency to the platform, and their swimming routes were significantly decreased, as compared with mice treated with DG alone. In addition, the combined supplementation (EGT+MEL) significantly decreased the total length of swimming routes, as compared to MEL or EGT supplementation alone.

Water-maze retention test. On day 60, the swimming time and total swimming distance in R2 quadrants (where the hidden platform was originally placed but now removed) in mice treated with DG alone were significantly reduced, as compared with those of control mice (Figure 4). Supplementation with EGT or MEL did not improve the memory retention in mice treated with DG. However,

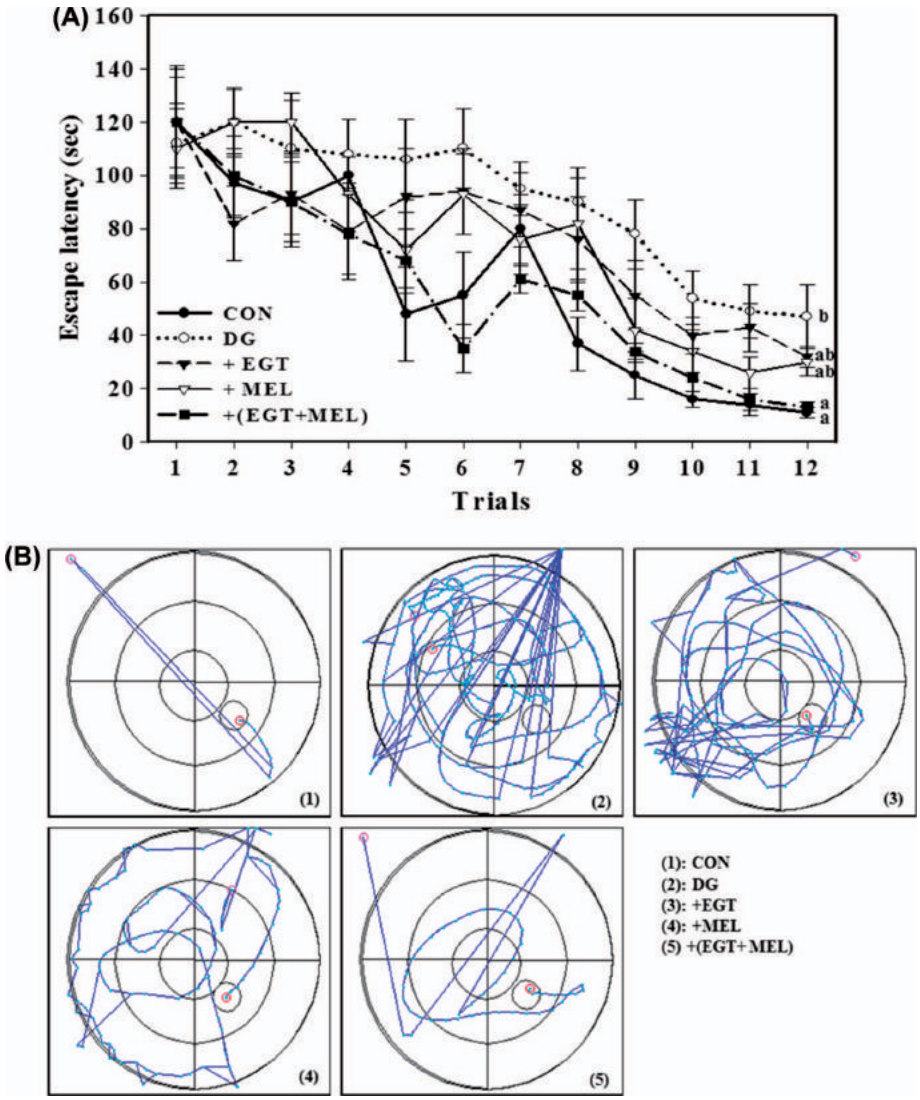

Figure 3. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on learning and memory ability in the water-maze memory acquisition test in mice treated s.c. with D-galactose (DG). The DG-treated mice were supplemented p.o. with EGT (0.5 mg/kg/d), MEL (10 mg/kg/d), or EGT+MEL. (A) Escape latency (i.e., time spent on finding the hidden platform placed in R2 quadrant) measured on days 57–59. (B) The swimming pathway of mice in the acquisition test measured on day 59. The small circle points the initial place where the mouse was put into the pool. The large circle points the location of hidden platform. The traces of swimming pathway that are outside of the computerized circle in Figure 5B are still within the rim of the water maze tank (not shown). Data (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different ($P < 0.05$).

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

supplementation of mice with EGT+MEL significantly increased the swimming time and distance to cross the R2 quadrants, as compared with those of mice treated with DG alone (Figure 4) ($P < 0.05$). The results indicate that MEL+EGT ameliorated the impairment of memory and learning ability of mice induced by DG.

Water-maze reacquisition tests. The reacquisition test was conducted on days 61 and 86. In this test, the hidden platform was moved from R2 quadrants to R3 quadrants and submerged into the water (0.4 cm) of the maze. In the formal test, the time to reach hidden platform (or escape latency) for mice treated with DG alone was significantly increased on both day 61 and day 86 (Figure 5). Supplementation with either EGT or MEL significantly decreased the escape latency, as compared with DG treatment alone on days 61 and 86. Furthermore, the combined supplementation (EGT+MEL) strongly decreased the escape latency on day 86, as compared with EGT and MEL supplementation alone, and the effect of combined supplementation appeared to be additive or even synergistic.

Effect of MEL and EGT on Aβ accumulation in hippocampus of DG-treated mice

Figure 6 shows that DG treatment alone markedly increased the number and percentage of Aβ in hippocampus of mice, as compared to the control group. Importantly, supplementation with EGT or MEL alone completely prevented the Aβ accumulation in the hippocampus, and thus no further improvement was achieved by the combined supplementation (EGT+MEL) in Aβ plaque formation.

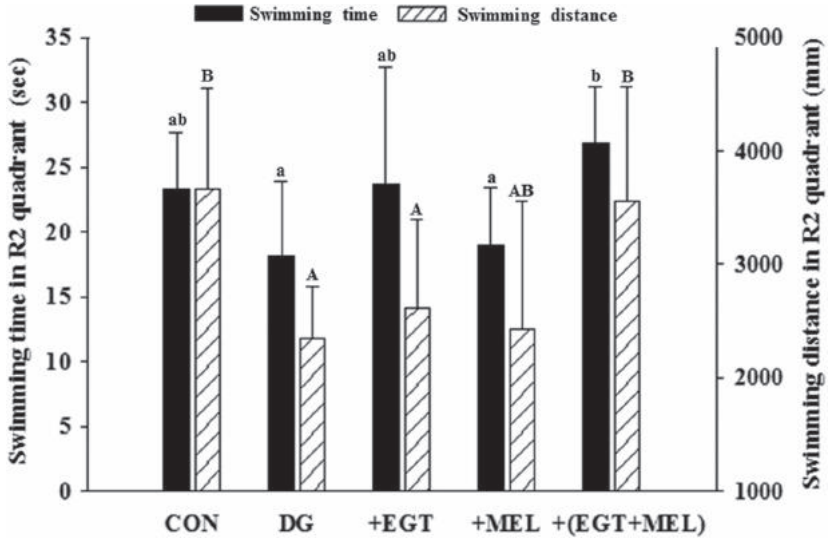

Figure 4. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on learning and memory ability in the water-maze memory retention test (on day 60) in mice treated s.c. with D-galactose (DG). The DG-treated mice were supplemented p.o. with EGT (0.5 mg/kg/d), MEL (10 mg/kg/d), or EGT+MEL. The hidden platform was removed in the retention tests, and mice were allowed to swim for 60 s in the water maze. The swimming route for each mouse in the pool was recorded, and the swimming time and total swimming distance at the R2 quadrant without the hidden platform were calculated. Data (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different (lower case for swimming time and upper case for swimming distance; $P < 0.05$).

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Effect of EGT on EGT level and number of OCTN1 plaque in the brain of DG-treated mice

As shown in Figure 7, the EGT contents in the brain tissues of mice supplemented with EGT (161 ± 19 μmol/mg protein) or EGT+MEL (153 ± 24 μmol/mg protein) were significantly higher than those of DG treatment (128 ± 6 μmol/mg protein) and MEL-treated mice (134 ± 14 μmol/mg protein). There were no significant differences among the control, DG treatment alone, and the MEL-supplemented groups in brain EGT contents. Thus, MEL supplementation could not prevent the loss of brain EGT contents in mice treated with DG. In addition, the change in number of OCTN1 plaque in brain tissues of mice in all groups was similar to that of the EGT content. However, both the EGT content and number of OCTN1 plaque in brain tissues of mice supplemented with EGT+MEL were not significantly different from those of mice supplemented with EGT.

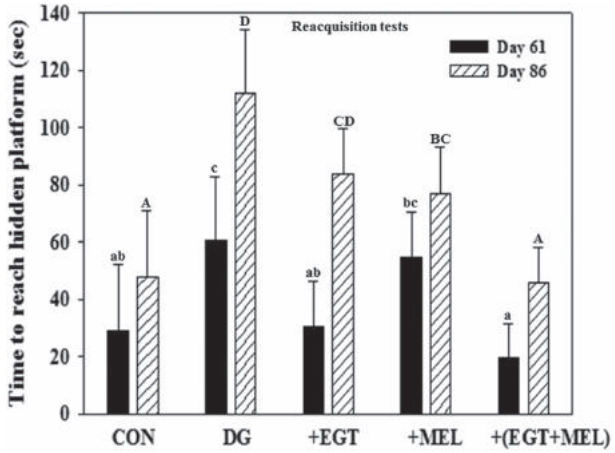

Figure 5. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on learning and memory ability in the water-maze memory reacquisition test (on days 61 and 86) in mice treated s.c. with D-galactose (DG). The DG-treated mice were supplemented p.o. with EGT (0.5 mg/kg/d), MEL (10 mg/kg/d), or EGT+MEL. Swimming time to reach the hidden platform (in R3 quadrant) in the reacquisition tests was measured. Data (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different (lower case for day 61 and upper case for day 86; $P < 0.05$).

Effect of EGT on antioxidant status and AChE activity in the brain of DG-treated mice

As also shown in Table I, treatment with DG alone significantly increased lipid peroxidation in the brain measured as TBARS. Supplementation with either EGT or melatonin completely inhibited DG-induced lipid peroxidation, and thus no further improvement could be achieved by the combined supplementation (EGT+MEL). In contrast, DG treatment alone markedly decreased the GSH/GSSG ratio (from 9.4 ± 1.3 to 5.0 ± 1.5), whereas supplementation with EGT, MEL, or EGT+MEL completely prevented the decline of GSH/GSSG ratio induced by DG. Surprisingly, the GSH/GSSG ratios in mice supplemented with EGT, MEL, or EGT+MEL were significantly higher than those in control mice, although no significant differences existed among the three supplemental groups. DG treatment alone also markedly decreased brain SOD activity (from 3.7 ± 0.7 unit/mg protein/min in the control group to 2.4 ± 0.4

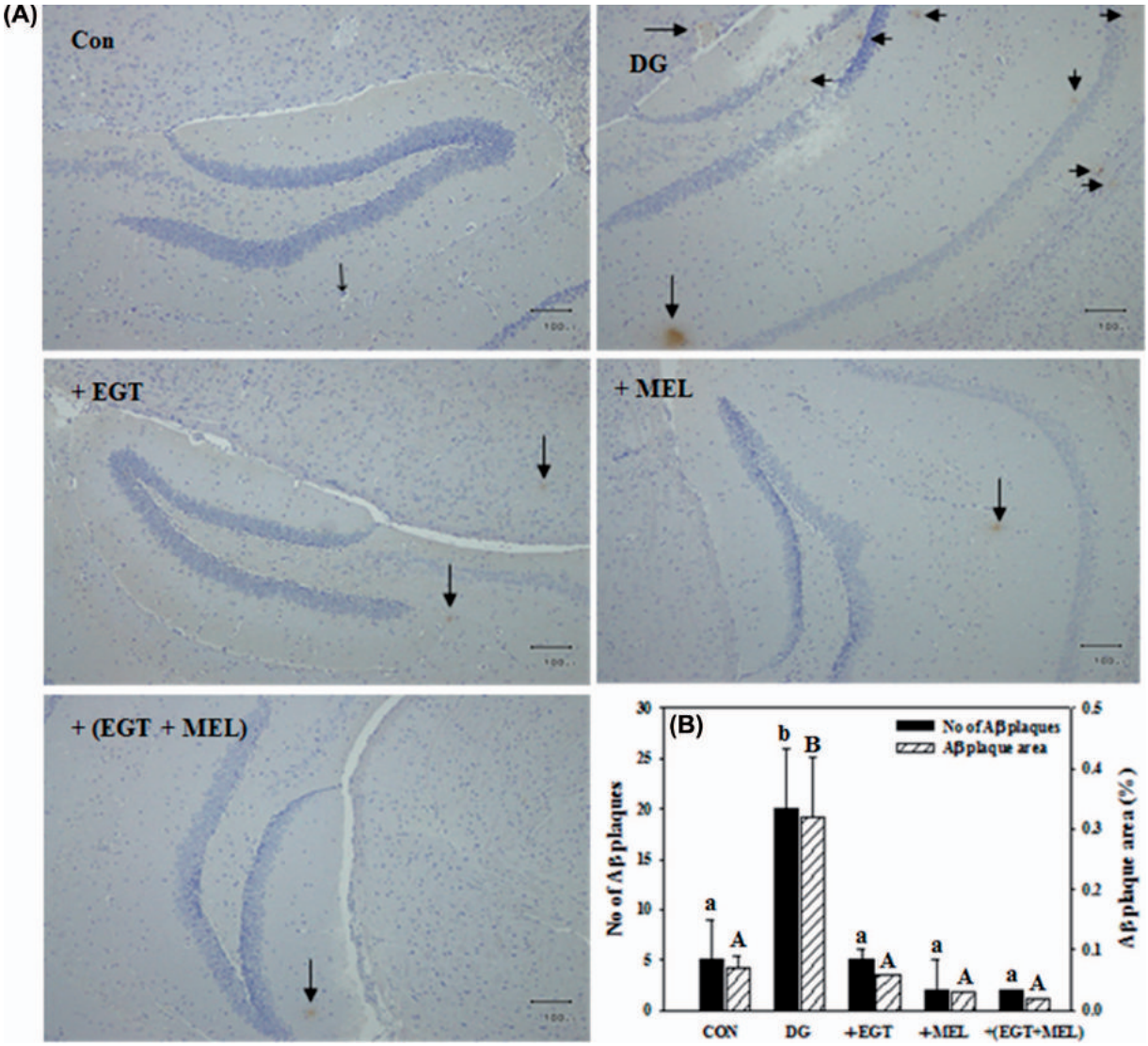

Figure 6. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on Aβ accumulation in the hippocampus (indicated with arrows) of mice treated s.c. with D-galactose (DG). The DG-treated mice were supplemented p.o. with MEL (10 mg/kg/d), EGT (0.5 mg/kg/d), or EGT+MEL. Immunohistochemical stain was carried out using the non-biotin hydrogen peroxidase kit. (A) The Aβ accumulation in hippocampus monitored by microscopic examination (×100) was shown by brown dye (big arrow head), and the nucleus of the section stained with hematoxylin was shown by blue dye (small arrow head). (B) Quantification of the accumulation of Aβ plaques in the hippocampus of mice by microscope automation and digital image system (Image Pro Plus 6.2). Values (means ± SD, $n = 9$ for the test groups) not sharing an alphabetic letter are significantly different (lower case for number of Aβ plaques and upper case for Aβ plaque area; $P < 0.05$).

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

unit/mg protein/min). Supplementation with EGT, MEL, or EGT+MEL tended to restore brain SOD activity, but the SOD activities in mice of these three supplementation groups were not significantly different from those of DG treatment group.

In mice treated with DG alone, the brain AChE activity was significantly increased (approximately 1.5-fold in comparison with the control group) (Table I). Supplementation with MEL completely prevented the rise in brain AChE activity induced by DG ($P < 0.05$), whereas supplementation with EGT only partially prevented the rise in brain AChE activity. The combined supplementation (EGT+MEL) did not further decrease the brain AChE activity, as compared to EGT or MEL supplementation alone.

Discussion

DG has been shown to induce oxidative stress and aging both *in vitro* and *in vivo* [39,40], while oxidative stress is known to induce Aβ formation and vice versa [41]. The main purpose of the present study was to evaluate possible combinational effects of EGT and MEL on learning and memory abilities using the DG induction model for AD in C57BL/6 mice. In addition, this study also served to substantiate the validity of DG induction model for AD in mice because it was unclear whether DG may induce learning and memory deficits in mice, although we have previously shown that DG induces oxidative stress and changes in some biochemical parameters related to aging and AD in the same strain of mice [30]. Data obtained

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

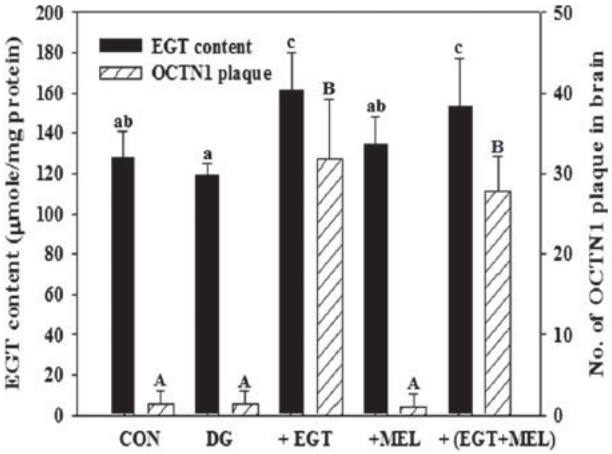

Figure 7. Effect of ergothioneine (EGT), melatonin (MEL), or their combination on brain EGT content and number of organic cation transporter-1 (OCTN1) plaques in mice treated s.c. with D-galactose (DG). The DG-treated mice were supplemented p.o. with EGT (EGT, 0.5 mg/kg/d), melatonin (MEL, 10 mg/kg/d), or EGT+MEL. Values (means ± SD, $n = 9$ mice per group) not sharing an alphabetic are significantly different (lower case for EGT content and upper case for number of OCTN1 plaque; $P < 0.05$).

from the present study confirm the reliability of DG induction model for AD, as DG induced behavioral retrogression such as the decrease of learning and memory ability in the active avoidance test and the water maze tests (acquisition, retention, and reacquisition tests). DG also induced a marked increase in Aβ accumulation, lipid peroxidation, and AChE activity as well as a decrease in the GSH/GSSG ratio and SOD activity in the brain of mice treated with DG. These results suggest that DG impairs learning and memory ability of the mice possibly through causing oxidative damage to the brain.

Using the DG induction model for studying oxidative damage in relation to AD, we observed that both EGT and MEL effectively ameliorated the behavioral performance in both active avoidance and the water-maze tasks and enhanced antioxidant status in DG-treated mice. Importantly, the combination of EGT and MEL appeared to exhibit synergistic neuroprotective effects on learning and memory activity. Studies have shown that MEL, the main hormone of the pineal gland that has been described to be a potent antioxidant [16,17], is lower in AD patients than in age-matched control subjects [18,19]. It is generally accepted that MEL is a potentially useful agent in attenuating and/or delaying AD. Jang et al. [13] have reported that EGT protects PC12 cells against Aβ-induced apoptosis by decreasing ROS and reactive nitrogen species, and they speculated that EGT used in conjunction with other antioxidants may be beneficial for prevention and treatment of AD. Here, we found additive or even synergistic effects of EGT and MEL on learning and memory, possibly due to their antioxidant activities leading to decreased oxidative stress and Aβ accumulation in hippocampus.

The water-maze test is a well-validated method for examining spatial learning and reversal learning in rodents [42]. We found a significant prolongation of escape latency in acquisition test in mice treated with DG (group 2), as compared to that in control group, indicating that the ability of learning was weakened by DG (Figure 5). DG treatment also impaired memory retention capacities, as both the swimming time and total swimming distance around the missing platform (originally at R2 quadrant) were significantly decreased (Figure 6). However, oral administration of EGT and MEL improved DG-induced deficits in memory retention, and the combination of MEL and EGT further improved these deficits. The water-maze reacquisition test (working memory test) determines whether the mice have developed new strategies to research the hidden platform in a novel situation (i.e., the hidden platform had been moved from R2 to R3 quadrant). We found that even after long period of resting (from day 61 to day 86), DG-treated mice still searched the platform without an obvious direction and could hardly reach the platform. However, supplementation with either EGT or MEL markedly decreased the time to reach the hidden platform on both days 61 and day 86. These results indicate that the information storage and cognitive flexibility are disrupted severely by DG treatment, which not only suppressed previously acquired behavior strategies but also hindered the ability to establish new strategies. Importantly, the combination of EGT and MEL further enhanced the protective effects on learning and memory activity over those of EGT and MEL individually.

Table I. Effects of supplementation of ergothioneine (EGT), melatonin (MEL), or their combination on antioxidant status and acetylcholinesterase (AChE) activity in the brain tissue of mice treated with D-galactose (DG).

Groups	TBARS (nmole/mg protein)	GSH/GSSG ratio	SOD (unit/mg protein/min)	AChE (nmol/5 min/mg protein)
Control	8.0 ± 0.4^{a}	9.4 ± 1.3^{b}	3.7 ± 0.7^{b}	34.1 ± 3.5^{a}
DG	9.5 ± 0.5^{c}	5.0 ± 1.5^{a}	2.4 ± 0.4^{a}	49.5 ± 5.1^{c}
+EGT	7.9 ± 0.4^{a}	10.2 ± 1.5^{c}	3.4 ± 1.1^{ab}	40.5 ± 4.2^{b}
+MEL	8.5 ± 0.5^{ab}	11.2 ± 3.4^{c}	3.4 ± 0.7^{ab}	34.1 ± 3.2^{a}
+(EGT+MEL)	8.1 ± 0.5^{a}	11.1 ± 1.9^{c}	3.1 ± 0.5^{ab}	39.8 ± 5.1^{b}

Values (means ± SD, $n = 9$ for the test groups) in each column not sharing the same superscript letter are significantly different ($P < 0.05$). TBARS, thiobarbituric acid-reactive substances; GSH/GSSG, glutathione/glutathione disulfide; SOD, superoxide dismutase.

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

Supplementation EGT and MEL alone or their combination also effectively inhibited lipid peroxidation in brain tissues. Lipid peroxidation can result in reducing membrane fluidity, inactivating membrane-bound proteins, and changing the structure and function of the membranes [43]. Supplementation with EGT and MEL alone or their combination completely prevented the decrease in GSH/GSSG ratio induced by DG. Strikingly, EGT and MEL alone or their combination increased the ratio to a level that was significantly higher than that in the control mice. DG also significantly decreased brain SOD activity, which was restored to some extent, albeit not significantly, by treatment with EGT, MEL, or EGT+MEL. As expected, DG treatment significantly increased the number of Aβ plaque in the hippocampus. However, supplementation with EGT and MEL individually or their combination completely prevented the accumulation of Aβ. The results suggest that the neuroprotective effects of EGT and/or MEL in this DG-induced mouse model are attributed, at least partially, to attenuation of oxidative stress and inhibition of Aβ formation.

The change of cholinergic system is well known for its critical role in learning and memory both in hippocampus and neocortex of mammalian [44]. AChE activity has been shown to be increased within and around amyloid plaques [45,46], to promote the assembly of amyloid beta-peptides into fibrils [47], and to increase the cytotoxicity of these peptides [48]. Here, we observed that the AChE activity was increased in the brain of DG-treated mice (Table I). As indicated by Kaizer et al. [49], the elevation of AChE activity may be due to a direct neurotoxic effect on the plasmatic membrane caused by increased lipid peroxidation, and that such changes may affect the integrity and functionality of the cholinergic system. Our observation that EGT was capable of preventing the increase of AChE activity suggests that EGT may play an important role in the maintenance of acetylcholine synaptic levels by preventing the increase of AChE activity. Thus, the normalization of AChE activity may be another mechanistic action by which EGT prevents learning and memory deficits and even improves these behavioral functions *in vivo*.

OCTN1 is functionally expressed in mouse brain, small intestine, liver, and kidney [50–52] and is a powerful and highly specific transporter for the uptake of EGT in humans and animals [53]. Cells lacking OCTN1 do not accumulate EGT, since the plasma membrane is virtually impermeable for this hydrophilic zwitterion compound. OCTN1 may transport EGT into neurons and consequently reduce the intracellular ROS level, resulting in suppression of neuronal proliferation [9]. Thus, dysfunction of OCTN1 may contribute to the development of neurodegenerative diseases, and therapeutic drugs and other compounds that target OCTN1 may influence neuronal differentiation or proliferation. Here, we show that the EGT supplementation increased both EGT and OCTN1 levels in brain tissues of mice, indicating that EGT can enter brain tissues and produce neuroprotective effects.

Conclusion

This study has demonstrated that supplementation with EGT, MEL, or their combination significantly improves memory and learning abilities in mice treated with DG. Importantly, the combination of EGT and MEL appeared to exhibit synergistic neuroprotective effects on learning and memory activity. This effect of EGT and/or MEL is likely through attenuation of oxidative damage, which could inhibit Aβ accumulation and prevent the rise in AChE activity in brain tissues of DG-induced mice. Further investigations are warranted to explore the possible involvement of other neurotransmitters, such as glutamate, GABA, and catecholamines [54], responsible for the memory-improving effect of combination of EGT and MEL.

Declaration of interest

The authors report no declarations of interest. The authors alone are responsible for the content and writing of the paper.

This study was supported by a grant from the National Science Council, ROC. (NSC 97-2313-B-235-001-MY3) and was supported in part by the Ministry of Education, Taiwan, ROC under the ATU plan.

References

[1] Jewart RD, Green J, Lu CJ, Cellar J, Tune LE. Cognitive, behavioral, and physiological changes in Alzheimer disease patients as a function of incontinence medications. Am J Geriatr Psychiatry 2005;13:324–328.

[2] Arendash GW, Schleif W, Reazi-zadeh K, Jackson EK, Zacharia LC, Cracchiolo JR, et al. Caffeine protects Alzheimer's mice against cognitive impairment and reduces brain-beta amyloid production. Neuroscience 2006;142:941–952.

[3] Butterfield DA, Shelley TR, Newman F, Sultana R. Roles of amyloid β peptide-associated oxidative stress and brain protein modifications in the pathogenesis of Alzheimer's disease and mild cognitive impairment. Free Radic Biol Med 2007;43:658–677.

[4] Rammouz G, Lecanu L, Aisen P, Papadopoulos V. A lead study on oxidative stress-mediated dehydroepiandrosterone formation in serum: the biochemical basis for a diagnosis of Alzheimer's disease. J Alzheimers Dis 2011;24:5–16.

[5] Schubert D, Behl C, Lesley R, Brack A, Dargusch R, Sagara Y, Kimura H. Amyloid peptides are toxic via a common oxidative mechanism. Proc Natl Acad Sci U S A 1995;92:1989–1993.

[6] Townsend KP, Pratico D. Novel therapeutic opportunities for Alzheimer's disease: focus on nonsteroidal anti-inflammatory drugs. FASEB J 2005;19:1592–1601.

[7] Butterfield DA, Boyd-Kimball D. Amyloid beta-peptide (1-42) contributes to the oxidative stress and neurodegeneration

Downloaded by [University of Sussex Library] at 10:05 24 September 2017

found in Alzheimer's disease brain. Brain Pathol 2004;14: 426–432.

[8] Melville DB, Horner WH, Otken CC, Ludwig ML. Studies on the origin of ergothioneine in animals. J Biol Chem 1955;213:61–68.

[9] Nakamichi N, Taguchi T, Hosotani H, Wakayama T, Shimizu T, Sugiura T, et al. Functional expression of carnitine/organic cation transporter OCTN1 in mouse brain neurons: possible involvement in neuronal differentiation. Neurochem Int 2012;61:1121–1132.

[10] Aruoma OI, Whiteman M, England TG, Halliwell B. Antioxidant action of ergothioneine: assessment of its ability to scavenge peroxynitrite. Biochem Biophys Res Commun 1997;231:389–391.

[11] Cheah IK, Halliwell B. Ergothioneine; antioxidant potential, physiological function and role in disease. Biochim Biophys Acta 2012;1822:784–793.

[12] Moncaster JA, Walsh DT, Gentleman SM, Jen LS, Aruoma OI. Ergothioneine treatment protects neurons against N-methyl-D-aspartate excitotoxicity in an *in vivo* rat retinal model. Neurosci Lett 2002;328:55–59.

[13] Jang JH, Aruoma OI, Jen LS, Chung HY, Surh YJ. Ergothioneine rescues PC12 cells from b-amyloid-induced apoptotic death. Free Radic Biol Med 2004;36:288–299.

[14] Song TY, Chen CL, Liao JW, Ou HC, Tsai MS. Ergothioneine protects against neuronal injury induced by cisplatin both *in vitro* and *in vivo*. Food Chem Toxicol 2010;48:3492–3499.

[15] Aruoma OI, Spencer JP, Mahmood N. Protection against oxidative damage and cell death by the natural antioxidant ergothioneine. Food Chem Toxicol 1999;37:1043–1053.

[16] Tan DX, Chen D, Poeggeler B, Manchester LC, Reiter RJ. Melatonin: a potent endogenous hydroxyl radical scavenger. Endocrinol J 1993;1:57–60.

[17] Tan DX, Hardeland R, Manchester LC, Poeggeler B, Lopez- Burillo S, Mayo JC, et al. Mechanistic and comparative studies of melatonin and classic antioxidants in terms of their interactions with the ABTS cation radical. J Pineal Res 2003;34:249–259.

[18] Liu RY, Zhou JN, van Heerikhuize J, Hofman MA, Swaab DF. Decreased melatonin levels in postmortem cerebrospinal fluid in relation to aging, Alzheimer's disease, and apolipoprotein E-epsilon4/4 genotype. J Clin Endocrinol Metab 1999;84:323–327.

[19] Ozcankaya R, Delibas N. Malondialdehyde, superoxide dismutase, melatonin, iron, copper, and zinc blood concentrations in patients with Alzheimer disease: cross-sectional study. Croat Med J 2002;43:28–32.

[20] Rosales-Corral SA, Acuña-Castroviejo D, Coto-Montes A, Boga JA, Manchester LC, Fuentes-Broto L, et al. Alzheimer's disease: pathological mechanisms and the beneficial role of melatonin. J Pineal Res 2012;52:167–202.

[21] Yang NC, Lin HC, Wu JH, Ou HC, Chai YC, Tseng CY, et al. Ergothioneine protects against neuronal injury induced by β-Amyloid in mice. Food Chem Toxicol 2012;50: 3902–3911.

[22] Holden HM, Rayment I, Thoden JB. Structure and function of enzymes of the Leloir pathway for galactose metabolism. J Biol Chem 2003;278:43885–43888.

[23] Cui X, Wang L, Zuo P, Han Z, Fang Z, Li W, Liu J. D-Galactose-caused life shortening in Drosophila melanogaster and Musca domestica is associated with oxidative stress. Biogerontology 2004;5:317–326.

[24] Deng H, Cui D, Jiang J, Feng Y, Cai N, Li D. Inhibiting effects of Achyranthes bidentata polysaccharide and Lycium barbarum polysaccharide on nonenzyme glycation in D-galactose induced mouse aging model. Biomed Environ Sci 2003;16: 267–275.

[25] Zhang Q, Li X, Cui X, Zuo P. D-Galactose injured neurogenesis in the hippocampus of adult mice. Neurol Res 2005;27:552–556.

[26] Lei H, Wang B, Li W, Yang Y, Zhou A, Chen M. Anti-aging effect of astragalosides and its mechanism of action. Acta Pharmacol Sin 2003;24:230–234.

[27] Song X, Bao MM, Li DD, Li YM. Advanced glycation in D-galactose induced mouse aging model. Mech Ageing Dev 1999;108:239–251.

[28] Ho SC, Liu JH, Wu RY. Establishment of the mimetic aging effect in mice caused by D-galactose. Biogerontology 2003;4:15–18.

[29] Lei M, Hua X, Xiao M, Ding J, Han Q, Hu G. Impairments of astrocytes are involved in the D-galactose-induced brain aging. Biochem Biophys Res Commun 2008;369: 1082–1087.

[30] Hsieh HM, Wu WM, Hu ML. Soy isoflavones attenuate oxidative stress and improve parameters related to aging and Alzheimer's disease in C57BL/6J mice treated with D-galactose. Food Chem Toxicol 2009;47:625–632.

[31] Anand KV, Mohamed SMJ, Thomas PA, Geraldine P. Protective role of chrysin against oxidative stress in D-galactose-induced aging in an experimental rat model. Geriatr Gerontol Int 2012;12:741–750.

[32] Cambell MD, Jaques Jr S, Gala RR. The lack of an effect of cholinergic agonists on anterior pituitary prolactin production *in vitro*. Experientia 1978;34:1522–1523.

[33] Doody RS. Clinical profile of donepezil in the treatment of Alzheimer's disease. Gerontology 1999;45:23–32.

[34] Lee CL, Kuo TF, Wang JJ, Pan TM. Red mold rice ameliorates impairment of memory and learning ability in intracerebroventricular amyloid β-infused rat by repressing amyloid β-accumulation. J Neurosci Res 2007;85:3171–3182.

[35] Schofield JD, Chen X. Analysis of free reduced and free oxidized glutathione in wheat flour. J Cereal Sci 1995;21: 127–136.

[36] Buege AJ, Aust SD. Microsomal lipid peroxidation. Methods Enzymol 1978;52:302–310.

[37] Ellman GL, Courtney KD, Andres VJ, Featherstone RM. A new and rapid colorimetric determination of acetylcholinesterase activity. Biochem Pharmacol 1961;7:88–95.

[38] Dubost NJ, Beelman R, Peterson D, Royse D. Identification and quantification of ergothioneine in cultivated mushrooms using liquid chromatography–mass spectroscopy. Int J Med Mushr 2007;8:215–222.

[39] Liu YY, Nagpure BV, Wong PT, Bian JS. Hydrogen sulfide protects SH-SY5Y neuronal cells against d-galactose induced cell injury by suppression of advanced glycation end products formation and oxidative stress. Neurochem Int 2013;62: 603–609.

[40] Chen B, Zhong Y, Peng W, Sun Y, Hu YJ, et al. Increased mitochondrial DNA damage and decreased base excision repair in the auditory cortex of D-galactose-induced aging rats. Mol Biol Rep 2011;38:3635–3642.

[41] Borza LR. A review on the cause-effect relationship between oxidative stress and toxic proteins in the pathogenesis of neurodegenerative diseases. Rev Med Chir Soc Med Nat Iasi 2014;118:19–27.

[42] D'Hooge R, De Deyn PP. Applications of the Morris water maze in the study of learning and memory. Brain Res Rev 2001;36:60–90.

[43] Li XM, Ma YL, Liu XJ. Effect of the Lycium barbarum polysaccharides on age-related oxidative stress in aged mice. J Ethnopharmacol 2007;111:504–511.

[44] Rogers JL, Kesner RP. Cholinergic modulation of the hippocampus during encoding and retrieval. Neurobiol Learn Mem 2003;80:332–342.

[45] Ulrich J, Meier-Ruge W, Probst A, Meier E, Ipsen S. Senile plaques: staining for acetylcholinesterase and A4 protein: a comparative study in the hippocampus and entorhinal cortex. Acta Neuropathol 1990;80:624–628.

[46] Morán MA, Mufson EJ, Gómez-Ramos P. Colocalization of cholinesterases with β amyloid protein in aged and Alzheimer's brains. Acta Neuropathol 1993;85:362–369.

[47] Inestrosa NC, Alvarez A, Pérez CA, Moreno RD, Vicente M, Linker C, et al. Acetylcholinesterase accelerates assembly of amyloid β-peptides into Alzheimer's fibrils: possible role of the peripheral site of the enzyme. J Neuron 1996;16: 881–891.

[48] Alvarez A, Alarcon R, Opazo C, Campos EO, Munoz FJ, Calderon FH, et al. Stable complexes involving acetylcholinesterase and amyloid-β peptide change the biochemical properties of the enzyme and increase the neurotoxicity of Alzheimer's fibrils. J Neurosci 1998;18:3213–3223.

[49] Kaizer RR, Corrêa MC, Spanevello RM, Morsch VM, Mazzanti CM, Gonçalves JF, Schetinger MR. Acetylcholinesterase activation and enhanced lipid peroxidation after long-term exposure to low levels of aluminum on different mouse brain regions. J Inorg Biochem 2005;99:1865–1870.

[50] Lamhonwah AM, Hawkins CE, Tam C, Wong J, Mai L, Tein I. Expression patterns of the organic cation/carnitine transporter family in adult murine brain. Brain Dev 2008;30:31–42.

[51] Sugiura T, Kato S, Shimizu T, Wakayama T, Nakamichi N, Kubo Y, et al. Functional expression of carnitine/organic cation transporter OCTN1/SLC22A4 in mouse small intestine and liver. Drug Metab Dispos 2010;38:1665–1672.

[52] Kato Y, Kubo Y, Iwata D, Kato S, Sudo T, Sugiura T, et al. Gene knockout and metabolome analysis of carnitine/organic cation transporter OCTN1. Pharm Res 2010;27:832–840.

[53] Gründemann D, Stephanie H, Golz S, Geerts A, Lazar A, Berkels R, et al. Discovery of the ergothioneine transporter. Proc Natl Acad Sci USA 2005;102:5256–5261.

[54] Parle M, Dhingra D, Kulkarni SK. Neurochemical basis of learning and memory. Ind J Pharm Sci 2004;66:371–376.

Downloaded by [University of Sussex Library] at 10:05 24 September 2017